MILITARY SPECIFICATION

# ELEMENT, RELAY, XM— LOADING, ASSEMBLING AND PACKING

## 1. SCOPE

1.1 This specification covers the quality assurance provisions of the loading, assembling and packing for one type of relay designated as Element, Relay, XM— (see 6.4).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

### SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

MIL-D-45608 — Designs and Drawings, Ordnance Inspection Equipment.

MIL-C-45662 — Calibration of Standards.

### STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1167 — Ammunition Data Cards.

MIL-STD-1168 — Lot Numbering of Ammunition.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-414 — Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

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DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8838191 — Element, Relay XM— Assembly.

7548389 — Box, Set-Up, Packing, Ammunition, for Relays and Primers.

7548347 — Box, Packing, Ammunition, for Relay, M4, M5, M7, XM10 or Primer, M42.

7548388 — Packing Box.

PUBLICATIONS

ORDNANCE CORPS

IEL–8838191 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

**3.1 Material.** Material and parts shall be in accordance with applicable drawings and specifications.

**3.2 Relays.** The relays shall comply with all requirements specified on Drawing (dwg) 8838191, and with all requirements specified in applicable specifications.

**3.3 Moisture content of lead azide.** The moisture content of the lead azide at the loading station at the time of loading the relays shall not exceed 0.50 percent, when tested as specified in 4.3.1.

**3.4 Workmanship.** All parts shall be free of chips, dirt, grease and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 General quality assurance provisions.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define the terms used herein. The provisions of Specification MIL–A–2550 shall apply.

**4.1.1** *Submission of product.* At the time of the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.



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(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificates shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the cerifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

**4.2 Inspection provisions.**

4.2.1 *Lot formation.* A lot shall consist of relay elements, produced by one manufacturer in one unchanged process in accordance with the same drawing, same drawing revision, same specification, and same specication revision. Drawing, specification and process changes not affecting safety, performance or fit, as determined by the Government, shall not be cause for changing the lot interfix number. The examination and tests listed herein shall be performed on inspection lots submitted in accordance with Standard MIL–STD–105, MIL–STD–1235, or MIL–STD–414, as applicable. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL–STD–1168. Each lot shall contain:

(a) Metal parts of one manufacturer bearing the same lot interfix number.

(b) Lead azide from one lot.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects when listed shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL or 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Once agreement has been reached as to which method will be used ( group or individual) that method will be used for all of the CD's listed in this specification. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Assembly (see dwg. 8838191).

| *Categories* | *Defects* | *Method of inspection* | *Code No. (see 6.2)* |
|---|---|---|---|
| Critical: | | | |
| 1. | Cup cracked or split | Visual | 01001* |

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| *Categories* | *Defects* | *Method of inspection* | *Code No. (see 6.2)* |
|---|---|---|---|
| Major: | AQL 0.40 percent | | |
| 101. | Charge missing | Visual | 01002 |
| 102. | Weight of charge (see 4.2.3.2) | Balance | 01003 |
| Minor: | AQL 2.50 percent | | |
| 201. | Outside diameter | Gage | 01004 |
| 202. | Total length | Gage | 01005 |
| 203. | Underside of crimped metal fails to contact disc 360 degrees | Visual | 01006 |
| 204. | Onion skin disc mutilated | Visual | 01007 |
| 205. | Evidence of poor workmanship (see 3.4) | Visual | 01008 |

**4.2.2.2** Box, set-up, inner (see dwg. 7548389).

| *Categories* | *Defects* | *Method of inspection* | *Code No.* |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.40 percent | | |
| 101. | Contents loose | Manual | 02001 |
| Minor: | AQL 0.65 percent | | |
| 201. | Marking missing, misleading or unidentifiable | Visual | 02002 |

**4.2.2.3** Box, set-up, outer, in barrier bag (see dwg. 7548347).

| *Categories* | *Defects* | *Method of inspection* | *Code No.* |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Seal improper or incomplete | Visual | 03001 |
| 102. | Bag punctured, torn, or cut | Visual | 03002 |
| 103. | Seal not permitting minimum subsequent seals | Visual | 03003 |
| 104. | Bag contains air (air not exhausted) | Manual | 03004 |
| Minor: | AQL 0.65 percent | | |
| 201. | Marking missing, misleading or unidentifiable | Visual | 03005 |

**4.2.2.4** Sealed wood packing box (see dwg. 7548388).

| *Categories* | *Defects* | *Method of inspection* | *Code No.* |
|---|---|---|---|
| Critical: | None defined. | | |



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| *Categories* | *Defects* | *Method of inspection* | *Code No.* |
|---|---|---|---|
| Major: | AQL 0.40 percent | | |
| 101. | Strapping missing, broken or loose | Visual-Manual | 04001 |
| 102. | Board broken or split | Visual | 04002 |
| Minor: | AQL 1.50 percent | | |
| 201. | Contents loose | Manual | 04003 |
| 202. | Strapping improperly assembled | Visual | 04004 |
| 203. | Marking misleading or unidentifiable | Visual | 04005 |

4.2.3 *Testing.*

4.2.3.1 *Moisture content of lead azide (see dwg. 8838191) — Major defect — Code No. 01009.* A sample shall be selected from each batch or lot of lead azide. It shall be of sufficient size for the performance of the determination in accordance with 4.3.1. Failure of the sample in meeting the requirement shall reject the batch or lot.

4.2.3.2 *Weight of charge (see dwg. 8838191)—Major defect—Code No. 01004.* Fifty unloaded cups shall be selected from each lot and shall be weighed separately. A sample of eighty (80) loaded units shall be weighed separately. The sum of the lightest weight of an empty cup and the minimum specified charge weight shall be used as the minimum acceptable weight for loaded units. Failure of two (2) or more units in meeting the minimum acceptable weight requirement shall reject the lot.

4.2.3.3. *Heat seal test (see dwg. 7548347) —Major defect—Code No. 03006.* A sample of 5 barrier bags shall be selected for this test from each lot of packaged relay elements and tested as specified on the applicable drawing, using test equipment in accordance with 4.2.4.

4.2.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL–8838191 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.2.4.4 *Calibration.* The contractor shall have available a set of master gages, standards, and appropriate instruments for regu-

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larly scheduled calibration of his acceptance equipment. Records of such regularly scheduled calibrations shall be maintained by the contractor and made available for review by the Government. The calibration of gages, standards, and instruments shall be in accordance with Specification MIL–C–45662.

**4.3 Test methods and procedures.**

**4.3.1** *Determination of moisture content of lead azide.* A tared, dried aluminum dish 2 inches in diameter shall be covered with a sheet of polytetrafluoroethylene 2½ inches square and sufficiently flat to form a tight cover. An accurately weighed portion of approximately two grams of the sample shall be transferred to an oven and dried at 95° plus or minus 2° Centigrade (C.) for 2 hours. The cover shall be placed in a sulfuric acid desiccator during the drying period. After drying, the dish shall be quickly transferred to the desiccator and allowed to cool to room temperature. The polytetrafluoroethylene cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated to percentage of moisture in the lead azide as follows:

$$\text{Percentage of moisture} = \frac{(A-B)\ 100}{A}$$

where:

A = weight of sample before drying.

B = weight of sample after drying.

## 5. PREPARATION FOR DELIVERY

**5.1 Packaging.**

**5.1.1** *Level C.—Unit pack.* The assemblies shall be unit packaged in accordance with dwg. 7548389.

**5.1.2** *Level C.—Intermediate pack.* Unit packages shall be intermediate packaged in accordance with dwg. 7548347.

**5.2 Packing.**

**5.2.1** *Level C.* The intermediate packages shall be packed in accordance with dwg. 7548388.

**5.3 Marking.** Marking shall be in accordance with dwg. 7548347, 7548388, or 7548389 as applicable.

## 6. NOTES

**6.1 Ordering data.** Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) *Data cards.* Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL–STD–1167.

**6.2 Inspection code numbers.** The five digit code number assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

**6.3 Inspection equipment.** The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraph 6.3.1 through 6.3.7.

**6.3.1** *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance



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with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

**6.3.2** *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF — Contractor's design responsibility on Army format in accordance with MIL–D–45608.

CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

**6.3.3** *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

**6.3.4** *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

**6.3.5** *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

**6.3.6** *Contractor designs.* Contractor designs are designs of inspection equipment for

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which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35 mm microfilm process. Contractors shall submit one copy of the final design as a rolled set or flat set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 *Submission of contractor designs.* Unless otherwise specified on the IEL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

**6.4 Intended use.** The Element, Relay, XM—, is used in Element, Delay or Non-Delay, M9 used with Bomb Fuzes, M904E2 and M905.

**6.5 Submission of critical defect.** For U.S. Army Munitions Command procurements, one copy of each critical defect report will be furnished the cognizant Government inspection agency; the Army Ammunition Procurement and Supply Agency, ATTN; SMUPA–RO; Picatinny Arsenal, ATTN; SMUPA–ND.

6.5.1 *Reporting to other services.* For other than U.S. Army Munitions Command procurements, critical defect reporting shall be as prescribed by the contracting officer.

**Custodian:**

**Army—MU**

**Preparing activity:**

**Army—MU**

**Project No. 1375–A554**



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